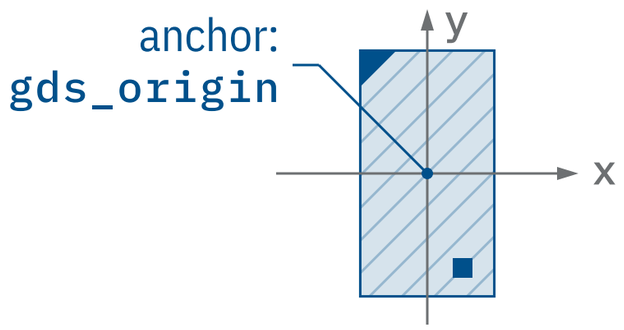
`global = position + R(rotation.z) * M * local`

die-local

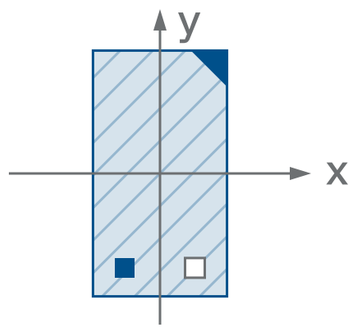


- pad at `(312.500, -845.000)` µm

the die's own GDS frame, y-up

step 1, mirror

`M = diag(-1, 1)` for orientation: `flip_chip`

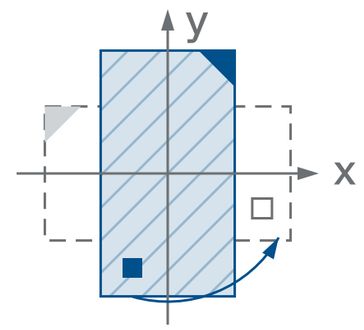


- `(-312.500, -845.000)`
- □ `M = diag(1, 1)` for `face_up`

only x changes sign; never mirror in y

step 2, rotate

`R(rotation.z)`



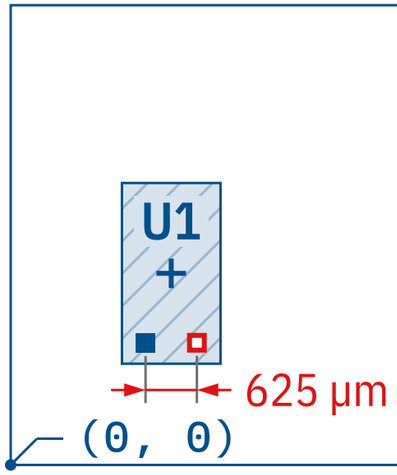
- `rotation.z = 0` `(-312.500, -845.000)`
- □ `rotation.z = 90` `(845.000, -312.500)`

step 3, translate by position

the order matters. Rotating before mirroring gives a different answer for every angle that is not a multiple of 180 degrees

position does not move when the die rotates, which is why the contract puts it at the geometric centre rather than at a corner

the canonical frame



a systematic offset of the whole pad field, not an obvious break; the GDS is well formed and the 3D view renders happily

- + U1 position `(1954.121, 2332.483)`
- ■ `flip_chip` result: `(1641.621, 1487.483)`
- □ if `face_up` were used by mistake: `(2266.621, 1487.483)`
- ⟷ 625 µm, more than eight Cu-pillar pitches at the 75 µm minimum